
BENJAMIN DOURTHE, PhD

AI Strategist & Machine Learning Leader | Digital Health & MedTech | Team & Program Builder

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SUMMARY

AI engineer specialized in building advanced harnesses that improve the performance, efficiency, and safety of agentic systems. Brings 12+ years across AI research, digital health, and MedTech, pairing hands-on depth in machine learning, computer vision, and signal processing (first-in-class wearable, implantable, and imaging systems taken from prototype to product) with enterprise leadership of generative and agentic AI: strategy, enablement, governance, and the tooling that makes AI-assisted development verifiable and validatable in a complex regulated environment. PhD-trained and research-grounded, known for fast, robust iteration, strong engineering discipline, and cross-functional leadership across software, engineering, clinical, and research teams.

CORE COMPETENCIES

Digital health & MedTech AI: biomedical sensing, medical imaging, digital biomarkers, wearables, implantables.

Agentic AI: platform rollout, prompt engineering, governance, security/compliance reviews.

Machine learning & deep learning: classification/regression, CNNs/RNNs, generative models, segmentation.

Computer vision & imaging: reconstruction, super-resolution (SRGAN), 2D/3D pipelines, quality optimization.

Time-series & signal processing: feature engineering, denoising, model robustness, generalization.

Production-grade Python: data pipelines, experiment tracking, code quality, automation, Git.

Leadership: hiring/interviewing, mentoring, roadmap delivery, stakeholder communication.

TECHNICAL SKILLS

Programming Languages: Python (primary), MATLAB, R, SQL, C, C++, Java, JavaScript, HTML/CSS.

Machine Learning & Deep Learning: scikit-learn, PyTorch, TensorFlow, hyperparameter optimization, model evaluation/validation, robustness testing, explainability.

Generative AI & Agentic Systems: Prompt and context engineering (advanced system prompts, reusable prompt frameworks), RAG design/build, org/team/project-specific skills; tools: Claude Code, OpenAI Codex, Gemini Enterprise, Cursor, Windsurf, GitHub Copilot, Deep Research.

Data, Signals & Imaging: NumPy, Pandas, SciPy, OpenCV, time-series analysis, feature engineering, multimodal data fusion, clinical-grade analytics.

DevOps, CI/CD & Engineering Practices: Git, testing, automation, CI/CD, documentation, reproducible research; architecture/branching strategies; data management for security/privacy/compliance; data & AI policy development.

Backend/Frontend & Full-Stack Foundations: APIs, backend and frontend frameworks, internal tooling and workflow automation for ops-heavy teams.

Domains: Digital health, MedTech, medical imaging, biomechanics, cardiac/pulmonary monitoring, wearables and sensor analytics; startup and early-stage product development.

PROFESSIONAL EXPERIENCE

STAFF MACHINE LEARNING ENGINEER, Supira Medical - Los Gatos, CA (2026 - Present)

MedTech company developing a next-generation percutaneous ventricular assist device designed to support cardiovascular function in high-risk patients.

- Lead the modernization of the Supira data platform by integrating new data streams, implementing advanced AI-powered functionalities, and automating manual steps to significantly accelerate processing speed, bolster security, and extract actionable insights from structured and unstructured data in preparation for pivotal clinical studies.
- Strengthen quality engineering by expanding automated test coverage and fortifying CI/CD pipelines. Transform verification and validation (V&V) from a manual effort into a repeatable, gated process that guarantees strict compliance for regulated pivotal-stage clinical programs.

- Direct company-wide AI enablement by managing agentic platform access and developing targeted training programs. Instruct rigorous cybersecurity assessments on new AI platforms to maintain secure, compliant usage guidelines across technical and regulatory teams.
- Collaborate with cross-functional teams to identify workflow bottlenecks and engineer custom automation or AI-driven solutions. Build specialized applications and guide departments in developing their own tools, directly increasing productivity, standardizing processes, and embedding engineering best practices across the organization.

STAFF MACHINE LEARNING ENGINEER, Shifamed / Adona Medical - Los Gatos, CA (2024 - 2026)

MedTech company developing adjustable cardiac shunt with implantable bi-atrial pressure monitoring systems.

- Directed the AI and Data Science operations and drove the enterprise rollout of generative and agentic AI systems (access management, policy, training, governance).
- Developed agentic data-processing and analytics pipelines plus custom team-specific applications across Software, EE, and R&D, and built an internal engineering toolkit containing hundreds of production-ready prompt and context-engineering templates.
- Engineered and validated signal-processing and machine learning algorithms for complex cardiac pressure data to support next-generation clinical tools and data infrastructure.

SENIOR MACHINE LEARNING ENGINEER, Nuralogix Corporation - Toronto, Canada (2022 - 2023)

Digital health company using computer vision for video-based physiological monitoring.

- Led end-to-end design and optimization of signal-processing and ML pipelines, improving video data quality by 25% and biometric prediction performance by 12% and engineered 100+ features for high-dimensional data representation and introduced robustness-focused evaluation and generalization strategies for real-world, multi-modal health data.

CHIEF DATA OFFICER, Tidal Medical - Toronto, Canada (2020 - 2022)

Startup developing first-in-class non-invasive wearable for real-time cardiopulmonary monitoring.

- Defined the data/AI architecture for a first-in-class non-invasive wearable for real-time pulmonary artery pressure monitoring, and pioneered deep learning for medical imaging enhancement (including SRGAN-based super-resolution) and redesigned signal-processing and computer-vision pipelines, improving algorithm efficiency by ~90% and enabling faster prototyping and more robust deployments.
- Built and led multidisciplinary teams, drove IP strategy and execution (1 PCT + 4 US provisional filings), and secured >\$2M in non-dilutive funding across national programs and grants.

MACHINE LEARNING ENGINEER (Postdoctoral), University of British Columbia - Vancouver, Canada (2020 - 2022)

- Designed and deployed deep learning models for medical image segmentation (multi-scale architectures) in collaboration with Medtronic and clinical stakeholders, improving accuracy and workflow efficiency for real-world clinical research.

COMPUTER VISION ENGINEER (Postdoctoral), Sunnybrook Research Institute - Toronto, Canada (2018 - 2020)

- Developed marker-less human motion analysis using deep learning and computer-vision, improving tracking accuracy by 35% and integrating multiple sensing modalities for real-time biomechanical analysis in research and clinical settings.

RESEARCH ENGINEER (PhD), KU Leuven - Belgium (2014 - 2017)

- Built patient-specific computational models and image-processing algorithms combining CT imaging with biomechanical simulation for 3D morphological feature extraction, stress mapping, and surgical planning.

EDUCATION

- **PhD, Biomedical Engineering** - KU Leuven, Belgium (2014 - 2017)
- **MSc, Science, Technology and Health** - University of Lorraine, France (2012 - 2013)
- **MSc, Mechanical Engineering** - National Engineering School of Metz (ENIM), France (2008 - 2013)
- *Visiting Scholar* - University of British Columbia, Canada (Fall 2016); University of Southern California, USA (Fall 2015)

SELECTED RESEARCH & FEATURED PROJECTS

- **Nexus-Hub (2026)** - an advanced harness designed to significantly enhance the capacities, usability and performance of agentic AI platforms (Claude Code, Codex, Cursor).
- **"The Future of Intelligence" (2025)** - 47-page independent essay analyzing the limitations of current AI architectures (LLMs, Transformers) and proposing research directions toward systems with genuine understanding. Includes mathematical foundations of attention mechanisms, predictive coding, and world models. <https://doi.org/10.5281/zenodo.18177869>
- **40+ publications & presentations** in biomedical engineering and applied machine learning (200+ citations).